

The P2 determinant as reverse Cauchy–Schwarz

Rank-1 identity, contribution anatomy, and the coarse bound that dies at one window

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Abstract

On a real Hermitian invertible matrix A_0 with exactly one negative eigenvalue $\lambda_- < 0$ and a two-column factor $U = (u_1, u_2)$, the Haynsworth matrix $K_2 = I_2 + U^T A_0^{-1} U$ admits the exact reverse-Cauchy identity

$$\det K_2 = \det(K_{2,+}) - Q, \quad Q = p^T \operatorname{adj}(K_{2,+}) p,$$

where $K_{2,+} = I + R_+$ is the positive remainder (the $J^{-1} = I_2$ summand plus the positive-rest Gram) and $p = U^T \psi_- / \sqrt{|\lambda_-|}$. Equivalently $Q = \|p\|^2 + p^T \operatorname{adj}(R_+) p$. The sign $\det K_2 < 0$ is $Q > \det(K_{2,+})$, i.e. the rank-1 negative-mode update dominates the remainder Gram. A direction-free sufficient bound $\|p\|^2 > \lambda_{\max}(K_{2,+})$ holds on 44/45 of the nneg-1 family and is killed at the same window that killed the r375 Cauchy test (kz 46). The contribution anatomy is mixed (no single term carries $-\det K_2$). The $C_{\min}$ mode of the r407 intertwiner is ψ_- . Dropping ψ_- restores $\det > 0$; vacuous windows ($n_- = 0$) have no pole and $\det K_2 > 0$. The r377 dictionary $\det K_2 < 0 \iff h_N h_{N+1} < 0$ is the LDL reading of the same mixed 2×2 . What remains of (P2) is the dressed coupling $p^T K_{2,+}^{-1} p > 1$ of the last two dual CD columns against $\psi_{C_{\min}}$. Nothing here is evidence for or against the Riemann hypothesis.

Status. Lemmas 1–3 and 4 are proved (finite real algebra). Lemma 5 is the r377 dictionary, re-read as the sign of the post-cap LDL pair. Proposition 1 is a finite machine check on the 45 resolvable nneg-1 windows, not a cofinal law. The direction-free coarse bound is *not* a theorem of the family (kz 46). The original statement $\det K_2 < 0$ on the nneg-1 branch is reduced to the dressed overlap $p^T K_{2,+}^{-1} p > 1$, with source ingredient $\langle u_{\text{CD}}, \psi_{C_{\min}} \rangle$. No RH claim. No L^* claim. No $R^\dagger$ claim.

1 The object

Let A_0 be real symmetric and invertible, of size $m \geq 2$, with inertia $(m - 1, 1, 0)$: one simple negative eigenvalue $\lambda_- < 0$ and a positive definite complement. Write ψ_- for a unit eigenvector of λ_- , A_+^{-1} for the inverse of A_0 on $\psi_-^\perp$ (zero on $\operatorname{span}\{\psi_-\}$), and $U = (u_1, u_2) \in \mathbb{R}^{m \times 2}$. Set

$$K_2 = I_2 + U^T A_0^{-1} U.$$

In the dual CD construction of rounds 367–371 one has $A_0 = R_{N-3} - \frac{1}{2}I$ on the node set Y and U the last two dual Christoffel–Darboux columns. The Haynsworth identity (sorry-free in `RH/Haynsworth.lean`) converts $\operatorname{ind}_-(A_0) = 1$ together with $\det K_2 < 0$ into $A_0 + U U^T \succ 0$. The mixed 3×3 of round 369 is $J = \operatorname{diag}(1, 1, 1/\operatorname{den})$; the present 2×2 is the case $J = I_2$, so the J^{-1} contribution is exactly I_2 .

2 Exact reverse-CS identity

Write $w = U^T \psi_-$ and $p = w / \sqrt{|\lambda_-|}$, so $p \in \mathbb{R}^2$. The spectral calculus is $A_0^{-1} = \lambda_-^{-1} \psi_- \psi_-^T + A_+^{-1}$ with $\lambda_- < 0$, hence

$$U^T A_0^{-1} U = -pp^T + R_+, \quad R_+ = U^T A_+^{-1} U \succeq 0.$$

Therefore

$$K_2 = K_{2,+} - pp^T, \quad K_{2,+} = I_2 + R_+. \quad (1)$$

The I_2 is the J^{-1} summand; R_+ is the positive-rest Gram.

Lemma 1 (Rank-1 determinant).

$$\det K_2 = \det(K_{2,+}) - Q, \quad Q = p^T \operatorname{adj}(K_{2,+}) p = a_+ p_2^2 - 2b_+ p_1 p_2 + d_+ p_1^2,$$

where $K_{2,+} = \begin{pmatrix} a_+ & b_+ \\ b_+ & d_+ \end{pmatrix}$. Equivalently $Q = \det(K_{2,+}) p^T K_{2,+}^{-1} p$, and

$$Q = \|p\|^2 + p^T \operatorname{adj}(R_+) p. \quad (2)$$

On the rational 4-node model $A_0 = \operatorname{diag}(-1, 1, 2, 3)$, $U = [(2, 0, 0, 0)^T, (1, 1, 0, 0)^T]$ one has $p = (2, 1)$, $K_{2,+} = \operatorname{diag}(1, 2)$, $\det(K_{2,+}) = 2$, $Q = 9$, $\det K_2 = -7$, and (2) reads $5 + 4$.

Proof. The matrix-determinant lemma on (1) is $\det(K_{2,+} - pp^T) = \det(K_{2,+}) (1 - p^T K_{2,+}^{-1} p)$. The 2×2 identity $\det(K_{2,+}) p^T K_{2,+}^{-1} p = p^T \operatorname{adj}(K_{2,+}) p$ is the first display. For the split, $\operatorname{adj}(I + R_+) = I + \operatorname{adj}(R_+)$ as 2×2 matrices, which is (2). The rational model is direct arithmetic (companion script). $\square$

Lemma 2 (Sign carrier). Assume $K_{2,+} \succ 0$ (true once $R_+ \succeq 0$). Then

$$\det K_2 < 0 \iff Q > \det(K_{2,+}) \iff p^T K_{2,+}^{-1} p > 1.$$

The first factor $\det(K_{2,+}) = \det(I + R_+) \geq 1$ cannot change sign (r375, SATZ).

Proof. Lemma 1 plus $\det(K_{2,+}) > 0$. $\square$

Lemma 3 (Direction-free sufficient bound). If $\|p\|^2 > \lambda_{\max}(K_{2,+})$, then $\det K_2 < 0$.

Proof. $Q = p^T \operatorname{adj}(K_{2,+}) p \geq \lambda_{\min}(\operatorname{adj}(K_{2,+})) \|p\|^2 = \det(K_{2,+}) \|p\|^2 / \lambda_{\max}(K_{2,+})$. The hypothesis forces $Q > \det(K_{2,+})$. $\square$

The bound is sufficient, not necessary. It is the reverse-CS form of the r375 Cauchy test $\|w\|^2 > |\lambda_-| (1 + \lambda_{\max}(R_+))$.

3 The $C_{\min}$ mode

Round 407 gives the intertwiner $R = C(I + C)^{-1}$, hence $A_0 = \frac{1}{2}(C - I)(C + I)^{-1}$ and $A_0^{-1} = 2(C + I)(C - I)^{-1}$. The two matrices are rational functions of each other, so they share eigenspaces.

Lemma 4 ($C_{\min}$ is ψ_-). The bottom eigenvector of C is ψ_- . In particular $\operatorname{ind}_-(A_0) = 1$ if and only if $n_C := \#\{\mu(C) < 1\} = 1$, and the coupling of the last two CD columns against the defect is

$$p_i = \frac{\langle u_i, \psi_{C_{\min}} \rangle}{\sqrt{|\lambda_-|}}.$$

Proof. Spectral calculus of the Cayley-type map $\lambda = (\mu - 1)/(2(\mu + 1))$: $\mu < 1$ iff $\lambda < 0$, and the eigenvectors coincide. Live cosine $1 - O(10^{-12})$ on every tested window (companion). $\square$

Both objects live at the cap: U is the pair of dual OP columns of degrees $N - 3, N - 2$, and $C_{\min}$ is the unique mode of the rank- $(N - 3)$ dual Gram that has crossed 1. The remaining arithmetic is a lower bound on that overlap, dressed by $K_{2,+}^{-1}$.

Gate protocol. Lemma 1–3 and 4 are unconditional finite algebra (no PNT, no RH). The named remainder $p^T K_{2,+}^{-1} p > 1$ is a source statement about dual CD columns versus the C -defect; it is not a prime-number theorem and is not an RH claim.

4 Anatomy, and the coarse kill

Proposition 1 (Branch check, not a law). *On the 45 resolvable nneg-1 windows of the r367 family (core-42 + r286-15 + EXT3/4; EXT5/6 are the sign-census class and were not rebuilt):*

- identity (1) holds at f64 grade (max residual $1.6 \cdot 10^{-10}$ on well-conditioned rows);
- $\det K_2 < 0$ on 45/45, $-\det K_2 \in [1.157, 1126.389]$, census floor 0.50 on 45/45;
- Rayleigh excess $p^T K_{2,+}^{-1} p - 1 \in [0.275, 3.295]$ on the core (min at kz 18); EXT excess ≥ 0.527 ;
- the direction-free bound of Lemma 3 holds 44/45 and fails at kz 46 ($\|p\|^2 = 2.852 < \lambda_{\max}(K_{2,+}) = 3.656$, alignment 0.984, $\det K_2 = -2.364$ still);
- $\|p\|^2 > \lambda_{\min}(K_{2,+})$ holds 45/45 with minimum gap 0.316 (kz 46) — a census inequality, not sufficient (the worst adj-direction still requires $\|p\|^2 > \lambda_{\max}$);
- the dominant summand of Q is mixed: $T_a = a_+ p_2^2$ on 25/45, $T_d = d_+ p_1^2$ on 4/45, $T_x = -2b_+ p_1 p_2$ on 16/45 (the cross term grows on EXT). No single-term bound covers the family;
- the J^{-1} share $\|p\|^2/Q$ lies in $[0.22, 0.59]$ on the core: the I_2 contribution alone does not dominate ($\|p\|^2 > \det(K_{2,+})$ on only 5/28 of the core);
- $\cos(\psi_-, \psi_{C_{\min}}) = 1$ and $n_C = 1$ on 45/45.

Fourteen of the forty-two core windows are vacuous ($n_- = 0$): there $K_{2,+} = K_2$, $\det K_2 > 0$ on 14/14, and $n_C = 0$. Overload $n_- \geq 2$ is 0/74.

The proposition is a machine check. The Reviewer’s hope that a coarse $O(1)$ bound on $\|p\|$ would close (P2) because $-\det K_2$ is huge is answered honestly: the *output* reserve is $O(1)$ and flat, but the direction-free input bound $\|p\|^2 > \lambda_{\max}(K_{2,+})$ is not universal on the family. The same window (kz 46) already killed the r375 Cauchy form; the alignment 0.984 is why (P2) still holds there.

5 Alternation

Lemma 5 (Post-cap dictionary). *On the nneg-1 branch, $\det K_2 < 0$ if and only if the product of the two LDL pivots of K_2 is negative. The r377 dictionary identifies those pivots, up to a positive Jacobi gauge, with (h_N, h_{N+1}) of the signed source, so*

$$\det K_2 < 0 \iff h_N h_{N+1} < 0.$$

This is not the r417 Woodbury-sch coordinate (r417: sch is a different 1×1).

The ψ_- -summand is the sign carrier: without it, $K_2 = K_{2,+} \succ 0$, both LDL pivots are positive, and $h_N h_{N+1} > 0$ (vacuous 29/29; explicit kz 12: $\det K_2 = +12.463$, first negative pivot at $N + 2$). On the branch the rank-1 update makes K_2 indefinite iff $Q > \det(K_{2,+})$, which is exactly the alternation. Live: 45/45 nneg-1 alternate; 29/29 vacuous do not.

6 Kills

mutant	hypothesis	$\det K_2$	note
vacuous $n_- = 0$ (14/42 core)	empty	> 0 (14/14)	no pole, $K_2 = K_{2,+}$
drop ψ_- at w9	$n_- = 1$ kept, pole omitted	+5.958	P2 dies
wrong $J = -I$ at w9	wrong remainder	+0.941	sign flips
dead χ_{3-15}	$n_- = 1$	-122.91	P2 <i>stays</i>
scramble w9	$n_- = 21$	-8.881	hypothesis empty
tiny overlap (toy)	$n_- = 1$, $\ p\ $ small	+1.0025	P1 $\nrightarrow$ P2

Dead character-twisted rungs on the nneg-1 branch keep $\det K_2 < 0$ (CHI3-15/19/39; CHI4-w9). Terminal death is a $q^\dagger$ statement (r433), not a P2 failure. The two coordinates are kept distinct from r417 sch.

7 Named remainder

Lemmas 1–2 convert (P2) into

$$p^T K_{2,+}^{-1} p > 1 \quad (\text{P2}^-)$$

with $p_i = \langle u_i, \psi_{C_{\min}} \rangle / \sqrt{|\lambda_-|}$. This is the r375 dressed overlap in reverse-CS letters, now with an identified source vector $\psi_{C_{\min}}$ and with a documented kill of every coarser direction-free bound that was tried. A numerical ε is not frozen: kz 18 has excess 0.275, compatible with any $\varepsilon \leq 1/4$, but a fitted constant is not a source statement.

The last two dual CD columns, against the unique C -mode below 1, must have $K_{2,+}$ -dressed overlap strictly larger than one. Expanding that overlap via the dual CD kernel on Y is the remaining finite identity. No $\lambda_{\min}$ of a Schur block of the full R , and no terminal margin, enters the target.

8 Claim boundary

Finite real algebra on a Hermitian pair (A_0, U) , plus a named remainder on the dual CD construction. Haynsworth remains sorry-free and does not assert (P1) or (P2) on windows. L^* , $R^\dagger \succeq \frac{1}{2}I$, and the Riemann hypothesis are not addressed in either direction.